EXPERIMENT 13

Case study on the limitations of low-code platforms for a complex enterprise use case.

Student details	Project details
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Department: CSBS	Platform: Zoho Creator (India region)
Institution: Rajalakshmi Engineering College	Evidence / verification date: 19 October 2026

Aim

To analyse the technical, operational and governance limitations of low-code development using an enterprise insurance-claims scenario.

Execution status

Case study completed using observed project behavior, platform documentation and clearly identified enterprise-scale assumptions. No unsupported enterprise load or compliance certification is claimed.

Application context

The project records insurance requests, validates submitted details, assigns claims for processing and presents claim status and amount summaries. Existing forms, reports, Deluge workflows and permission profiles provide the base for this experiment.

Screenshots in this report are actual captures from the project account. Explanations, tables and code listings are documentation, not simulated platform screenshots.

Enterprise case and limitations

Consider an insurer expanding this classroom application to many branches, policy systems, assessors and document sources. Claims may require multiple invoices, audit trails, asynchronous integrations and legally governed decisions. The current demonstration proves a smaller workflow, not this enterprise operating capacity.

Limitation	Project evidence and enterprise implication
AI reliability	Zia could suggest analysis but could not verify live counts and misunderstood an existing dashboard. Generated advice requires inspection and testing.
Query scale	An educational full scan matches aggregate counts on three records. This does not establish large-volume response time; selectivity and pagination matter.
Data modelling	The basic-field form does not implement repeating invoices, normalized payments or a claim-event history. These need explicit schema and retention design.
Platform abstraction	Indexing is platform-managed; runtime and API behavior require vendor documentation and support.
Identity and ownership	Own-record profiles require real authenticated identities. Public form submission alone cannot prove ownership for later status access.
External dependencies	Mail connections and future Catalyst services introduce authorization, error handling, quotas and operational ownership.

Risk analysis and mitigation

Risk / priority	Recommended response and acceptance test
Unauthorized claim access / High	Test staff and policyholder accounts independently, including direct URLs and APIs. Require denied cross-user access.
Duplicate integrations / High	Introduce idempotency keys and retry tracking; replay the same event and confirm one external side effect.
Poor response time / Medium	Use selective queries and bounded UI lists; benchmark realistic record volumes and document p50/p95 latency.
Missing audit evidence / High	Track status actor, timestamp and reason in an immutable or controlled history; verify completeness through a sample lifecycle.
Vendor dependency / Medium	Keep schema, source, API contracts and export/recovery procedures documented; test recovery in an isolated environment.
Configuration drift / Medium	Use staged releases and record versions; compare deployed behavior and access settings with the release checklist.

Architecture decision

Retain Creator for forms, workflow orchestration and staff interfaces. Use a custom service only when a measured requirement exceeds native capabilities. Keep insurance eligibility decisions under authorized human and organizational processes; the demonstration analytics function only summarizes data.

Cost and governance

Evaluate licenses, environment access, integration calls, storage, support and operational ownership before expansion. The observed trial-expiry notice is an account condition, not a benchmark of the platform or evidence of a purchased production service.

Evidence and conclusion

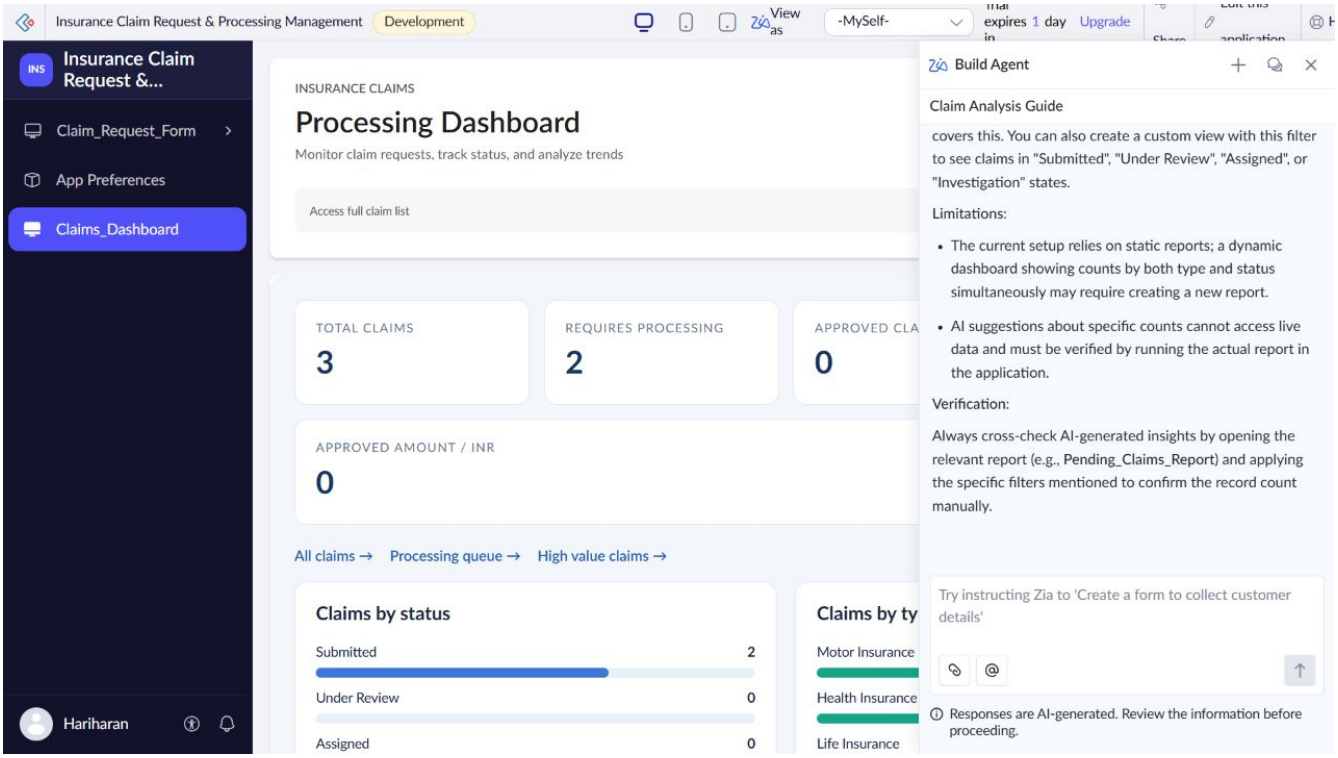


Figure 13.1 - Actual assistant limitations provide evidence for the AI reliability discussion.

Result

The case study identifies where low-code accelerates delivery and where enterprise requirements demand additional modelling, testing, governance or custom code. The application is a useful working prototype; enterprise readiness remains a separate engineering and organizational assessment.

Discussion questions

1. Why does correct output on three records not establish scalability?

2. How do authentication and own-record permissions differ from public form access?

3. When does an external service reduce complexity, and when does it increase operational risk?

References

Platform performance and limitations:
https://help.zoho.com/portal/en/kb/creator/developer-guide/others/platform-performance/articles/platform-performance